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#Name: Tanvi Verma
#Assignment 6: Python programming language
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#Problem 1: BUGGY CODE
#fix the above problem with super in python
class Person:

    def __init__(self, name, id):
        self.name = name
        self.id = id

    def Display(self):
        print("Name:", self.name)
        print("ID:", self.id)

class Emp(Person):
    def __init__(self, name, id):
        super().__init__(name, id)

    def Print(self):
        print("Employee class called")

Emp_details = Emp("Mayank", 103)

Emp_details.Display()

Emp_details.Print()
```

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Name: Mayank
ID: 103
Employee class called
```

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#problem 2 : Write a python program to find anagram check
def anagram_check(str1, str2):
    str1 = str1.replace(" ", "").lower()
    str2 = str2.replace(" ", "").lower()

    if sorted(str1) == sorted(str2):
        return True
    else:
        return False

str1 = input("Enter first string: ")
str2 = input("Enter second string: ")

if anagram_check(str1, str2):
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print("The strings are Anagrams.")  
else:  
    print("The strings are not Anagrams.")
```

Enter first string: listen
Enter second string: silent
The strings are Anagrams.

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#Problem 3: Write a python program to check for armstrong number  
num = int(input("Enter a number: "))  
original_num = num  
  
digits = len(str(num))  
sum = 0  
while num > 0:  
    digit = num % 10  
    sum = sum + digit ** digits  
    num = num // 10  
  
if sum == original_num:  
    print(original_num, "is an Armstrong Number.")  
else:  
    print(original_num, "is not an Armstrong Number.")
```

Enter a number: 153
153 is an Armstrong Number.

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#Problem 4: Deal with classes and objects  
class Vehicle:  
    def __init__(self, max_speed, mileage):  
        self.max_speed = max_speed  
        self.mileage = mileage  
        self.color = "White"  
        self.seating_capacity = 0  
  
    def assign_seating_capacity(self, capacity):  
        self.seating_capacity = capacity  
  
    def display_properties(self):  
        print("Maximum Speed:", self.max_speed, "kmph")  
        print("Mileage:", self.mileage, "kmpl")  
        print("Color:", self.color)  
        print("Seating Capacity:", self.seating_capacity)  
  
car1 = Vehicle(200, 20)  
  
car1.assign_seating_capacity(5)  
  
car2 = Vehicle(180, 25)  
  
car2.assign_seating_capacity(4)
```

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print("----- Car 1 Details -----")
car1.display_properties()

print()

print("----- Car 2 Details -----")
car2.display_properties()
```

```
----- Car 1 Details -----
Maximum Speed: 200 kmph
Mileage: 20 kmpl
Color: White
Seating Capacity: 5

----- Car 2 Details -----
Maximum Speed: 180 kmph
Mileage: 25 kmpl
Color: White
Seating Capacity: 4
```